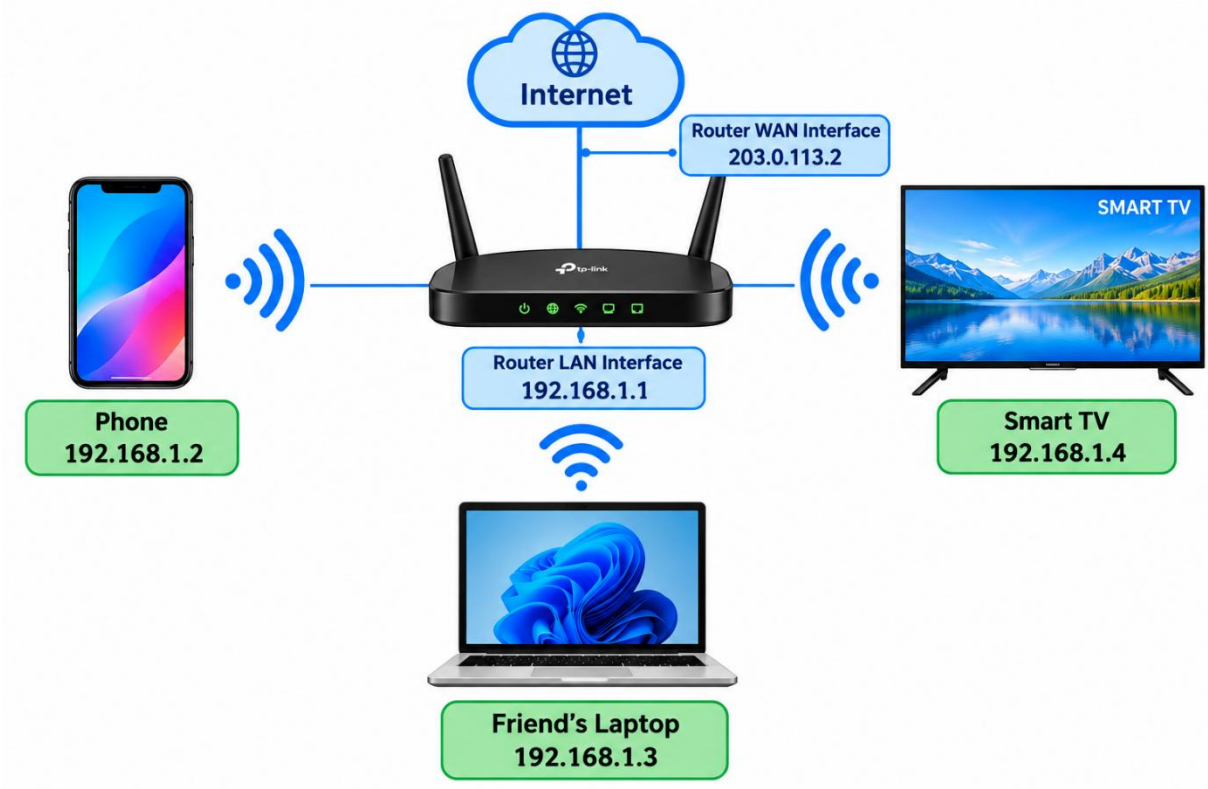
Assignment – 1.9

IP routing and routing protocols

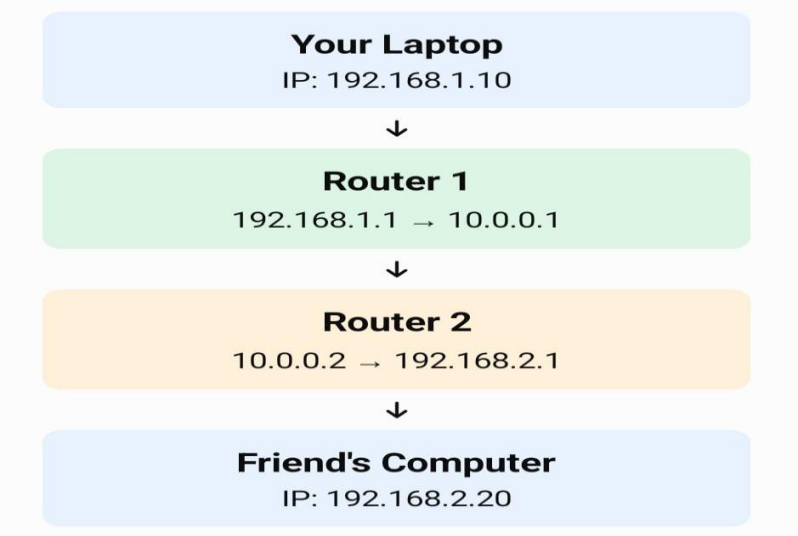
Q - 1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



Q - 2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

➢ Manual IP Routing:

Example: Sending a packet from your laptop to your friend's computer in a different subnet.



➢ Steps:

1. **Laptop:** Creates a packet with source IP 192.168.1.10 and destination IP 192.168.2.20.

2. **Routing table:** The laptop finds that the destination is in another subnet, so it sends the packet to the default gateway 192.168.1.1.

3. **Router 1:** Checks its routing table and forwards the packet to Router 2 at 10.0.0.2.

4. **Router 2:** Checks its routing table and finds that the 192.168.2.0/24 network is directly connected.

5. **Destination:** Router 2 forwards the packet to the friend's computer at 192.168.2.20.

6. **Reply:** The friend's computer sends a reply through its gateway, and the routers forward it back to your laptop.

➢ Routing Table Example:

Router	Destination network	Next hop
Laptop	192.168.2.0/24	192.168.1.1
Router 1	192.168.2.0/24	10.0.0.2
Router 2	192.168.2.0/24	Directly connected

Q - 3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

- **Routing Table – Zomato Delivery Network**

Network Address	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	Direct	Eth0
192.168.2.0	255.255.255.0	10.0.0.2	Eth1
192.168.3.0	255.255.255.0	10.0.0.3	Eth2

- **Branches:**
  - Ahmedabad: 192.168.1.0
  - Mumbai: 192.168.2.0
  - Pune: 192.168.3.0

Q - 4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

- **Static Routing vs Dynamic Routing:**

Routing	Pros	Cons
Static Routing	1. Easy to configure. 2. Uses fewer resources. 3. More secure.	1. Manual setup. 2. Hard to manage large networks. 3. Cannot automatically handle failures.
Dynamic Routing (RIP/OSPF)	1. Updates routes automatically. 2. Handles network failures. 3. Suitable for large networks.	1. Uses more resources. 2. Complex to configure. 3. Requires maintenance.

➢ Examples:

- **Static:** A small restaurant uses a fixed route to connect its billing system.
- **Dynamic:** A food delivery service uses dynamic routing to help network routers find alternative paths when a connection fails.

<u>Q - 5.</u> Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. Hint: Paste the AI-generated config and annotate each part with a short description.

- **<u>OSPF Configuration Between Two Routers:</u>**
- Router 1 (R1)

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

- router ospf 1 – Enables OSPF.
- network 192.168.1.0 – Shares R1's network.
- network 10.0.0.0 – Connects OSPF with R2.

- Router 2 (R2)

router ospf 1

network 192.168.2.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

- router ospf 1 – Enables OSPF.
- network 192.168.2.0 – Shares R2's network.
- network 10.0.0.0 – Connects OSPF with R1.